CLOUD SECURITY QUESTION BANK

UNIT 4

Section A - MCQs –1 Marks

1. What approach would you use in the ITIL lifecycle when a company prepares to roll out a new IT service and needs to design it?
   a) Service Strategy
   b) Service Design
   c) Service Transition
   d) Service Operation
   Answer: b) Service Design

2. What approach would you use to keep systems secure by regularly updating and fixing vulnerabilities?
   a) Patch Management
   b) Data Backup
   c) Encryption
   d) Firewalls
   Answer: a) Patch Management
3. What would you do if a cloud service provider receives negative performance feedback and must apply corrections to improve service delivery?
   a) Service Transition
   b) Service Operation
   c) Continual Service Improvement
   d) Service Strategy
   Answer: c) Continual Service Improvement
4. What examples can you find of data protection methods to secure information in cloud computing?
   a) Encryption
   b) Key management
   c) Access control
   d) All of the above
   Answer: d) All of the above

5. What approach would you use if a hospital wants to check its cloud services and possible risks before migration?
   a) Identification and assessment
   b) Monitoring
   c) Encryption
   d) Patch management
   Answer: a) Identification and assessment

Section B - Fill in the blanks – 1 Marks

1. What approach would you use if you wanted to categorize cloud services into SaaS, PaaS, and IaaS delivery models?
   Answer: SPI Model

2. How would you use IAM architecture to control user access in the cloud?
   Answer: Identification, Authentication, Authorization

3. What approach would you use if a cloud provider scans systems to detect weaknesses before hackers exploit them?
   Answer: Vulnerability management

4. How would you improve cloud security management when migrating data to the cloud?
   Answer: Minimize risks from threats and vulnerabilities

5. What would you do if an organization wants to clearly define responsibilities between the customer and the cloud provider?
   Answer: Apply the Shared Responsibility Model

Section C- Short Answer Questions – 3 Marks

1. What approach would you use to apply security management standards in the cloud?
   Ans:
   - Identify & assess all cloud services in use.
   - Apply encryption and access controls to reduce risks.
   - Conduct regular security audits to monitor threats.

2. What focus areas would you apply to secure cloud services effectively?
   Ans:
   - Protect data at rest, in transit, and in storage.
   - Apply shared responsibility model between provider and customer.
   - Use encryption and proper key management to secure data.

3. Demonstrate how IAM can be applied for secure access in cloud?
   Ans:
   - **Identification**: Assign unique identity to users or entities.
   - **Authentication**: verifying the claimed identity to ensure the person or entity is who they say they are.
   - **Authorization**: Grant permissions based on verified identity.

4. How would you apply the Shared Responsibility Model in securing cloud services?
   Ans:
   - Defines division of responsibilities between provider and customer.
   - **Cloud provider** Secures the cloud infrastructure (data centers, servers, networks).
   - **Customer** Secures applications, data, access controls, and encryption.

5. How would you manage vulnerabilities and patches to protect cloud systems?

Ans

- **Vulnerability Management:** Detect and assess weak points in systems using tools like vulnerability scanners
- **Patch Management**: Security Patch Management ensures software stays updated by applying patches that fix vulnerabilities and protect systems from security breaches.
- Together, prevent attacks by securing applications and systems.

Section D - Long Answer Questions – 5 marks

1. How would you apply the ITIL lifecycle to manage IT services in an enterprise?

Ans:

ITIL (Information Technology Infrastructure Library) framework provides best practices to plan, deliver, and improve IT services to meet the needs of the organization and its customers.

The ITIL lifecycle has five interconnected stages:

1. **Service Strategy** – Decide what IT services are needed to achieve organizational goals.
2. **Service Design** – Create detailed plans and specifications to ensure services meet business needs.
3. **Service Transition** – Build, test, and introduce services into the live environment smoothly.
4. **Service Operation** – Manage services in the live environment as per agreed service levels.
5. **Continual Service Improvement** – Continuously evaluate and improve services to meet evolving needs.

Each stage connects to the next, forming a continuous cycle of planning, designing, transitioning, operating, and improving IT services.



2. What approach would you use to apply security management functions across the SPI cloud delivery models?

Ans:

SPI stands for **Service Provider Interface**, typically referring to the three cloud service models: **SaaS, PaaS, and IaaS**.

1. **Software as a Service (SaaS):**
   • User Authentication & Authorization – Secure login ensures only authorized

users access applications.
• Data Encryption – Protects sensitive data during transmission and storage.

2. **Platform as a Service (PaaS):**
   • Access Controls – Manage user permissions and restrict unauthorized access.
   • Secure Development Practices – Enforce secure coding to reduce vulnerabilities.
3. **Infrastructure as a Service (IaaS):**
   • Virtual Machine Security – Ensure VMs are patched, updated, and properly configured.
   • Data Backup & Disaster Recovery – Prevent data loss and ensure business continuity.
   • Identity & Access Management – Control access to infrastructure resources.

3. How would you ensure availability management in an IaaS environment?
Ans:

- IaaS Availability Management ensures that the infrastructure is consistently accessible and operational.
- Providers must maintain computing and storage availability along with services like account management, messaging, authentication, databases, billing, and monitoring.
- Customers must configure and manage their virtual servers using the provider's available network, servers, and storage.
- Providers ensure reliable data centre design to handle failures and keep services running.
- Thus, availability is a shared responsibility between provider and customer.

4. Demonstrate how you would apply standard practices as a PaaS customer to reduce software application vulnerability?
Ans:

As a PaaS customer, security practices can be applied in the following ways:

1. **Secure Key Management**:
   - Store keys and credentials in **Azure Key Vault** to keep them safe.
   - Azure Key Vault encrypts authentication keys, storage account keys, and other sensitive data, ensuring they remain secure.
2. **Avoid Source Code Secrets**:
   - avoid placing secrets or credentials in source code or repositories,
   - By keeping keys out of public repositories, can prevent unauthorized access and reduce security risks.
3. **Protect VM Management Interfaces:**
   - Avoid exposing VM management interfaces directly to the internet.
   - Instead, I would use secure protocols such as SSH, RDP, or PowerShell remoting to manage VMs safely.

Applying these practices helps minimize vulnerabilities and strengthens the overall security of PaaS applications.

5. How would you apply LDAP, SAML, and OpenID protocols to improve authentication and access management in organizations?

Ans:

1. **LDAP (Lightweight Directory Access Protocol):**
   - Used to authenticate and manage on-premise applications.
   - Example: Employees authenticate through Active Directory to securely access internal resources.
2. **SAML (Security Assertion Markup Language):**
   - Applied in **Single Sign-On (SSO)** environments, allowing users to log in once and access multiple applications without repeated logins.
   - Example: A user logs in once and gains access to **multiple SaaS apps** like Salesforce and Google Workspace, improving efficiency.
3. **OpenID:**
   - Provides a **single digital identity** across consumer web applications.
   - Example: A user's **Google ID** is used to log into YouTube and third-party apps, keeping the same profile across platforms.

Application: Together, these protocols streamline authentication, reduce repeated logins, and strengthen access security in both **enterprise** and **consumer environments**.